POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 5033A | Semester V | Theory | 4 Credits

Fundamentals of Control Systems

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Electrical & Electronics Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 5033A as Fundamentals of Control Systems in Semester V for Electrical & Electronics Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Madras’s Control Engineering course and polytechnic control system curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Control system designs, stability assessments, automation logic and PID settings must be based on approved engineering models, certified automation software, current safety standards and competent professional review.

Verified source note: Verified sources support system modeling, Laplace transforms, transfer functions, block diagrams, time response (1st/2nd order), Routh-Hurwitz, Bode plots and PID controllers. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Fundamentals of Control Systems studies the mathematical modeling and analysis of dynamic systems to achieve desired performance through feedback. It develops the ability to represent systems using transfer functions and block diagrams, analyze time and frequency responses, evaluate stability using Routh-Hurwitz and Bode plots, and design PID controllers while respecting the technical and safety boundaries of automated control systems.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Engineering mathematics (Laplace transforms), basic electrical engineering, circuit theory and elementary physics.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain control system fundamentals, including open-loop, closed-loop and feedback principles.
2. Explain mathematical modeling of systems using Laplace transforms and transfer functions.
3. Analyze system performance using block diagram reduction, signal flow graphs and time response analysis.
4. Evaluate system stability and frequency response using Routh-Hurwitz criterion, Bode plots and PID control design.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the concepts of feedback control and the mathematical modeling of dynamic systems.	Understand
CO2	Explain block diagram reduction and signal flow graph techniques for system representation.	Understand
CO3	Analyze the time response of first-order and second-order systems to standard test signals.	Analyze
CO4	Evaluate system stability and frequency response using classical control techniques.	Evaluate

Official CO–PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	System Modeling and Transfer Functions	Introduction to control systems (open-loop vs closed-loop); feedback principles; Laplace transforms and inverse transforms; differential equations of physical systems; transfer function of electrical and mechanical systems.	Approx. 14 hours	CO1	Module 1
2	System Representation and Reduction	Block diagram representation; block diagram reduction rules; signal flow graphs; Mason's gain formula; comparison of reduction techniques; modeling of DC motors and sensors.	Approx. 16 hours	CO2	Module 2
3	Time Response Analysis	Standard test signals (step, ramp, impulse); time response of first-order systems; time response of second-order systems; transient response specifications (rise time, peak time, overshoot, settling time); steady-state errors.	Approx. 16 hours	CO3	Module 3
4	Stability and Frequency Response	Concept of stability (BIBO); Routh-Hurwitz stability criterion; frequency response basics; Bode plots; gain margin and phase margin; introduction to P, I, D controllers and their effects.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

System Modeling and Transfer Functions

Introduction to control systems (open-loop vs closed-loop); feedback principles; Laplace transforms and inverse transforms; differential equations of physical systems; transfer function of electrical and mechanical systems.

CO1 · Approx. 14 hours

System Representation and Reduction

Block diagram representation; block diagram reduction rules; signal flow graphs; Mason's gain formula; comparison of reduction techniques; modeling of DC motors and sensors.

CO2 · Approx. 16 hours

Time Response Analysis

Standard test signals (step, ramp, impulse); time response of first-order systems; time response of second-order systems; transient response specifications (rise time, peak time, overshoot, settling time); steady-state errors.

CO3 · Approx. 16 hours

Stability and Frequency Response

Concept of stability (BIBO); Routh-Hurwitz stability criterion; frequency response basics; Bode plots; gain margin and phase margin; introduction to P, I, D controllers and their effects.

CO4 · Approx. 14 hours

MODULE 1

System Modeling and Transfer Functions

Introduction to control systems (open-loop vs closed-loop); feedback principles; Laplace transforms and inverse transforms; differential equations of physical systems; transfer function of electrical and mechanical systems.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

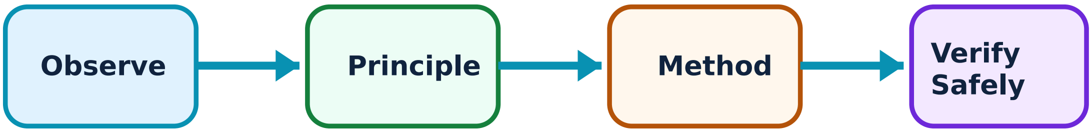
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for System Modeling and Transfer Functions: identify the system, state its principle, follow the correct method and verify the result safely.

1. Control system concepts−

Control systems manage the behavior of other devices or systems. Open-loop systems have no feedback, while closed-loop systems use the difference between the desired and actual output (error) to adjust the control action. Feedback is the core of modern automation, providing robustness and accuracy.

Study and application method

Compare open-loop and closed-loop systems with a conceptual example (e.g., a toaster vs an air conditioner); list three advantages of feedback.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare open-loop and closed-loop systems with a conceptual example (e.g., a toaster vs an air conditioner); list three advantages of feedback.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Feedback can cause instability if not properly designed. Do not implement high-gain feedback without authorised stability analysis.

2. Mathematical modeling−

Dynamic systems are modeled using differential equations. Laplace transforms convert these into algebraic equations in the 's-domain'. The transfer function is the ratio of the Laplace transform of the output to the input, representing the system's dynamic characteristics.

Study and application method

Derive the transfer function of a simple RC circuit conceptually; explain the significance of 'poles' and 'zeros' in a transfer function.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

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Common mistake / precaution: Models are approximations of real systems. Ensure that the model includes all safety-critical dynamics and accounts for parameter uncertainties.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

System Representation and Reduction

Block diagram representation; block diagram reduction rules; signal flow graphs; Mason's gain formula; comparison of reduction techniques; modeling of DC motors and sensors.

Approx. 16 hours | CO2 | Understand · Apply

Learning goals

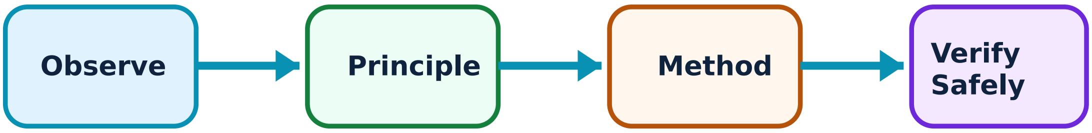
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for System Representation and Reduction: identify the system, state its principle, follow the correct method and verify the result safely.

1. Block diagrams and reduction−

Block diagrams visually represent the flow of signals and the functions of components. Complex systems are simplified using reduction rules (e.g., combining series/parallel blocks, moving summing points) to find the overall closed-loop transfer function.

Study and application method

Sketch a block diagram for a feedback system and simplify it using reduction rules to find C(s)/R(s).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a block diagram for a feedback system and simplify it using reduction rules to find C(s)/R(s).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Incorrect block diagram reduction leads to erroneous system analysis. All simplifications must be verified against the original system equations.

2. Signal flow graphs−

Signal flow graphs (SFG) provide an alternative graphical representation. Mason's gain formula allows for the direct calculation of the transfer function from the SFG without the step-by-step reduction required in block diagrams.

Study and application method

Explain the components of a signal flow graph (nodes, branches, loops); describe how Mason's gain formula is applied to find the system gain.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the components of a signal flow graph (nodes, branches, loops); describe how Mason's gain formula is applied to find the system gain.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: SFG analysis requires careful identification of all forward paths and loops. Overlooking a loop can lead to incorrect stability predictions.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Time Response Analysis

Standard test signals (step, ramp, impulse); time response of first-order systems; time response of second-order systems; transient response specifications (rise time, peak time, overshoot, settling time); steady-state errors.

Approx. 16 hours | CO3 | Understand · Apply

Learning goals

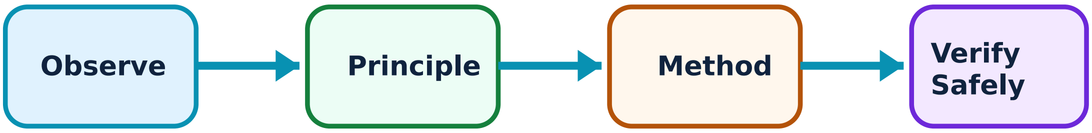
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Time Response Analysis: identify the system, state its principle, follow the correct method and verify the result safely.

1. Transient response specifications−

Transient response describes how a system moves from one state to another. For second-order systems, specifications like rise time, peak overshoot and settling time are critical for evaluating performance and ensuring the system reaches the target without excessive oscillation.

Study and application method

Sketch the unit step response of an underdamped second-order system and label the transient specifications.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the unit step response of an underdamped second-order system and label the transient specifications.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: High peak overshoot can damage mechanical components or saturate electronic circuits. All transient responses must stay within authorised safety limits.

2. Steady-state error analysis−

Steady-state error is the difference between the input and output as time approaches infinity. It depends on the system type (number of integrators) and the type of input signal. Minimizing steady-state error is essential for precision control.

Study and application method

Define 'system type' and calculate the steady-state error conceptually for a Type 1 system with a ramp input.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'system type' and calculate the steady-state error conceptually for a Type 1 system with a ramp input.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Zero steady-state error often requires high-gain integration, which can degrade stability. Balance error requirements with stability margins.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Stability and Frequency Response

Concept of stability (BIBO); Routh-Hurwitz stability criterion; frequency response basics; Bode plots; gain margin and phase margin; introduction to P, I, D controllers and their effects.

Approx. 14 hours **CO4** **Understand · Apply**

Learning goals

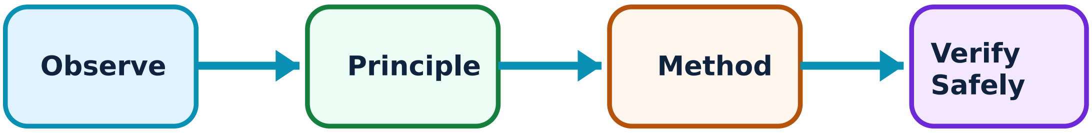
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Stability and Frequency Response: identify the system, state its principle, follow the correct method and verify the result safely.

1. Stability and Routh-Hurwitz−

A system is stable if every bounded input produces a bounded output (BIBO stability). The Routh-Hurwitz criterion determines stability by analyzing the coefficients of the characteristic equation without solving for the roots, identifying the number of poles in the right-half plane.

Study and application method

Explain the BIBO definition of stability; describe how to construct a Routh array for a given characteristic equation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the BIBO definition of stability; describe how to construct a Routh array for a given characteristic equation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Routh-Hurwitz only indicates absolute stability. Relative stability and performance must be evaluated using other techniques like Bode or Root Locus.

2. Frequency response and PID control−

Frequency response analysis (Bode plots) shows how the system responds to sinusoidal inputs of varying frequencies. Gain and phase margins indicate relative stability. PID (Proportional-Integral-Derivative) controllers are used to tune the system response for desired stability and error characteristics.

Study and application method

Sketch a conceptual Bode plot and identify the gain margin and phase margin; list the effects of increasing the 'Integral' gain (Ki) on steady-state error.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a conceptual Bode plot and identify the gain margin and phase margin; list the effects of increasing the 'Integral' gain (Ki) on steady-state error.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: PID tuning must be performed by competent personnel. Incorrect parameters can lead to violent oscillations or system failure. Use authorised tuning methods (e.g., Ziegler-Nichols).

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: System Modeling and Transfer Functions

Use the concept flow to revise observation, principle, method and verification.

Module 2: System Representation and Reduction

Use the concept flow to revise observation, principle, method and verification.

Module 3: Time Response Analysis

Use the concept flow to revise observation, principle, method and verification.

Module 4: Stability and Frequency Response

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare open-loop and closed-loop control in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the block diagram of a DC motor speed control system.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the transient specifications conceptually for a given second-order transfer function.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Determine the stability of a system conceptually using the Routh-Hurwitz criterion.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the gain and phase margins from a supplied Bode plot and state if the system is stable.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — System Modeling and Transfer Functions

Explain Control system concepts and state one correct method, application or precaution.−

Control systems manage the behavior of other devices or systems. Open-loop systems have no feedback, while closed-loop systems use the difference between the desired and actual output (error) to adjust the control action. Feedback is the core of modern automation, providing robustness and accuracy.

Answer framework: Compare open-loop and closed-loop systems with a conceptual example (e.g., a toaster vs an air conditioner); list three advantages of feedback.

Important point: Feedback can cause instability if not properly designed. Do not implement high-gain feedback without authorised stability analysis.

Explain Mathematical modeling and state one correct method, application or precaution.−

Dynamic systems are modeled using differential equations. Laplace transforms convert these into algebraic equations in the 's-domain'. The transfer function is the ratio of the Laplace transform of the output to the input, representing the system's dynamic characteristics.

Answer framework: Derive the transfer function of a simple RC circuit conceptually; explain the significance of 'poles' and 'zeros' in a transfer function.

Important point: Models are approximations of real systems. Ensure that the model includes all safety-critical dynamics and accounts for parameter uncertainties.

Module 2 — System Representation and Reduction

Explain Block diagrams and reduction and state one correct method, application or precaution.−

Block diagrams visually represent the flow of signals and the functions of components. Complex systems are simplified using reduction rules (e.g., combining series/parallel blocks, moving summing points) to find the overall closed-loop transfer function.

Answer framework: Sketch a block diagram for a feedback system and simplify it using reduction rules to find C(s)/R(s).

Important point: Incorrect block diagram reduction leads to erroneous system analysis. All simplifications must be verified against the original system equations.

Explain Signal flow graphs and state one correct method, application or precaution.−

Signal flow graphs (SFG) provide an alternative graphical representation. Mason's gain formula allows for the direct calculation of the transfer function from the SFG without the step-by-step reduction required in block diagrams.

Answer framework: Explain the components of a signal flow graph (nodes, branches, loops); describe how Mason's gain formula is applied to find the system gain.

Important point: SFG analysis requires careful identification of all forward paths and loops. Overlooking a loop can lead to incorrect stability predictions.

Module 3 — Time Response Analysis

Explain Transient response specifications and state one correct method, application or precaution.−

Transient response describes how a system moves from one state to another. For second-order systems, specifications like rise time, peak overshoot and settling time are critical for evaluating performance and ensuring the system reaches the target without excessive oscillation.

Answer framework: Sketch the unit step response of an underdamped second-order system and label the transient specifications.

Important point: High peak overshoot can damage mechanical components or saturate electronic circuits. All transient responses must stay within authorised safety limits.

Explain Steady-state error analysis and state one correct method, application or precaution.−

Steady-state error is the difference between the input and output as time approaches infinity. It depends on the system type (number of integrators) and the type of input signal. Minimizing steady-state error is essential for precision control.

Answer framework: Define 'system type' and calculate the steady-state error conceptually for a Type 1 system with a ramp input.

Important point: Zero steady-state error often requires high-gain integration, which can degrade stability. Balance error requirements with stability margins.

Module 4 — Stability and Frequency Response

Explain Stability and Routh-Hurwitz and state one correct method, application or precaution.−

A system is stable if every bounded input produces a bounded output (BIBO stability). The Routh-Hurwitz criterion determines stability by analyzing the coefficients of the characteristic equation without solving for the roots, identifying the number of poles in the right-half plane.

Answer framework: Explain the BIBO definition of stability; describe how to construct a Routh array for a given characteristic equation.

Important point: Routh-Hurwitz only indicates absolute stability. Relative stability and performance must be evaluated using other techniques like Bode or Root Locus.

Explain Frequency response and PID control and state one correct method, application or precaution.−

Frequency response analysis (Bode plots) shows how the system responds to sinusoidal inputs of varying frequencies. Gain and phase margins indicate relative stability. PID (Proportional-Integral-Derivative) controllers are used to tune the system response for desired stability and error characteristics.

Answer framework: Sketch a conceptual Bode plot and identify the gain margin and phase margin; list the effects of increasing the 'Integral' gain (Ki) on steady-state error.

Important point: PID tuning must be performed by competent personnel. Incorrect parameters can lead to violent oscillations or system failure. Use authorised tuning methods (e.g., Ziegler-Nichols).

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: System Modeling and Transfer Functions.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: System Representation and Reduction.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Time Response Analysis.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Stability and Frequency Response.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Introduction to control systems (open-loop vs closed-loop); feedback principles; Laplace transforms and inverse transforms; differential equations of physical systems; transfer function of electrical and mechanical systems..
2. Develop a complete answer or practical plan for: Block diagram representation; block diagram reduction rules; signal flow graphs; Mason's gain formula; comparison of reduction techniques; modeling of DC motors and sensors..
3. Develop a complete answer or practical plan for: Standard test signals (step, ramp, impulse); time response of first-order systems; time response of second-order systems; transient response specifications (rise time, peak time, overshoot, settling time); steady-state errors..
4. Develop a complete answer or practical plan for: Concept of stability (BIBO); Routh-Hurwitz stability criterion; frequency response basics; Bode plots; gain margin and phase margin; introduction to P, I, D controllers and their effects..

LAST-MILE REVISION

Quick Revision

Module 1: System Modeling and Transfer Functions

Introduction to control systems (open-loop vs closed-loop); feedback principles; Laplace transforms and inverse transforms; differential equations of physical systems; transfer function of electrical and mechanical systems.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: System Representation and Reduction

Block diagram representation; block diagram reduction rules; signal flow graphs; Mason's gain formula; comparison of reduction techniques; modeling of DC motors and sensors.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Time Response Analysis

Standard test signals (step, ramp, impulse); time response of first-order systems; time response of second-order systems; transient response specifications (rise time, peak time, overshoot, settling time); steady-state errors.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Stability and Frequency Response

Concept of stability (BIBO); Routh-Hurwitz stability criterion; frequency response basics; Bode plots; gain margin and phase margin; introduction to P, I, D controllers and their effects.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Control system concepts	Control systems manage the behavior of other devices or systems.
Mathematical modeling	Dynamic systems are modeled using differential equations.
Block diagrams and reduction	Block diagrams visually represent the flow of signals and the functions of components.
Signal flow graphs	Signal flow graphs (SFG) provide an alternative graphical representation.
Transient response specifications	Transient response describes how a system moves from one state to another.
Steady-state error analysis	Steady-state error is the difference between the input and output as time approaches infinity.
Stability and Routh-Hurwitz	A system is stable if every bounded input produces a bounded output (BIBO stability).
Frequency response and PID control	Frequency response analysis (Bode plots) shows how the system responds to sinusoidal inputs of varying frequencies.

LEARNING RESOURCES

References

Recommended control-systems references

1. I. J. Nagrath and M. Gopal, Control Systems Engineering.
2. K. Ogata, Modern Control Engineering.
3. Norman S. Nise, Control Systems Engineering.
4. B. C. Kuo, Automatic Control Systems.

Approved public control-systems sources

- https://nptel.ac.in/courses/108106098
- https://onlinecourses.nptel.ac.in/noc20_ee90/preview

These sources provide the verified study basis while official Course 5033A detail is unavailable; they do not replace official automation designs, authorised stability analysis, certified control software or authorised industrial plans.